

CyberEscape Room - Gamifying Cybersecurity Awareness

Problem Statement

Cybersecurity training is often passive and ineffective. Many individuals and organizations lack hands-on experience in identifying and mitigating cyber threats. We aim to change this with an engaging, gamified cybersecurity escape room.

Our Solution - CyberEscape Room

A virtual escape room where players solve security challenges to 'break out.' This immersive platform enables users to experience real-world cybersecurity threats in a safe and engaging environment.

Core Features

- AI-Powered Hints for players stuck on challenges.
- Encryption/Decryption puzzles.
- Web security challenges like SQL Injection & XSS.
- Simulated phishing attack detection.
- Multiplayer mode for team collaboration.

Technical Approach & Future Vision

Tech Stack

- AI & NLP: OpenAI API for AI hints
- Web Backend: Node.js & Firebase
- Security Tools: OWASP Zap, Burp Suite
- Game UI: React.js

Gameplay Flow

1. Players take on cybersecurity roles (Ethical Hacker, Analyst, etc.)
2. Solve real-world security challenges to progress.
3. Use AI hints if stuck, at the cost of points.
4. Multiplayer teams work together to solve threats.
5. Escape the virtual cyber-attack before time runs out!

Future Vision

Beyond the hackathon, CyberEscape Room can evolve into a full-fledged cybersecurity education platform for students, professionals, and organizations. It can integrate with cybersecurity certifications, making learning interactive and practical.